Documentation

Naresh Amitesh

June 2026

1 Introduction

The system designed implements an interest rate curve construction and swap pricing framework. It constructs discount curves using both cash instruments and swap instruments, supports linear and averaged quadratic interpolation, prices a new swap, calculates its par swap rate, and computes analytical risk vectors with respect to the input market quotes.

The implementation is structured so that curve construction, interpolation, pricing, and risk calculation are separated into independent components. This improves readability, maintainability, and extensibility.

2 System Functionality

The system contains functions that can perform the following tasks:

- Calculate discount factors from cash rates and par swap rates.
- Construct discount curves using both linear interpolation and averaged quadratic interpolation.
- Use the constructed curve to calculate the present value and par swap rate of a new swap.
- Calculate risk vectors analytically for each interpolation and curve construction method.

3 Mathematical Methodology

3.1 Discount Factors

For a cash quote c_i at maturity m_i, the discount factor is given by

$$DF(m_i) = \frac{1}{1 + \frac{c_i m_i}{360}}$$

3.2 Swap Present Value

For a fixed-rate payer swap, the present value is

$$PV = N\left(1 - DF(T)\right) - NK \sum_{j=1}^{M} \alpha_j DF(t_j)$$

where

- N is the notional,
- K is the fixed rate,
- T is the swap maturity,

- t_j are the fixed-leg payment dates,
- $\alpha_j = \frac{t_j - t_{j-1}}{360}$.

The corresponding par swap rate is

$$K_{\text{par}} = \frac{1 - DF(T)}{\sum_{j=1}^{M} \alpha_j DF(t_j)}$$

3.3 Cash Curve Risk

The risk with respect to cash quote c_i is defined as

$$R_i = \frac{\partial PV}{\partial c_i}$$

Differentiating the swap present value gives

$$R_i = -N \frac{\partial DF(T)}{\partial c_i} - NK \sum_{j=1}^{M} \alpha_j \frac{\partial DF(t_j)}{\partial c_i}$$

For the cash node,

$$\frac{\partial DF(m_i)}{\partial c_i} = -\frac{\frac{m_i}{360}}{\left(1 + \frac{c_i m_i}{360}\right)^2}$$

The interpolation method then determines

$$\frac{\partial DF(t_j)}{\partial DF(m_i)}$$

which is combined using the chain rule.

3.4 Linear Interpolation Risk

For $t \in [m_i, m_{i+1}]$,

$$DF(t) = DF(m_i)^{1-w} DF(m_{i+1})^{w}$$

where

$$w = \frac{t - m_i}{m_{i+1} - m_i}$$

Differentiating,

$$\frac{\partial DF(t)}{\partial DF(m_i)} = (1 - w) \frac{DF(t)}{DF(m_i)}$$

and

$$\frac{\partial DF(t)}{\partial DF(m_{i+1})} = w \frac{DF(t)}{DF(m_{i+1})}$$

3.5 Averaged Quadratic Risk

Let $Q_i(t)$ and $Q_{i+1}(t)$ denote the two quadratic interpolants used over the interval $[m_i, m_{i+1}]$.

The interpolation is given by

$$\log DF(t) = (1-w)Q_i(t) + wQ_{i+1}(t)$$

Therefore,

$$DF(t) = e^{(1-w)Q_i(t)+wQ_{i+1}(t)}$$

Differentiating,

$$\frac{\partial DF(t)}{\partial DF(m_k)} = DF(t)\left[(1-w)\frac{\partial Q_i(t)}{\partial DF(m_k)} + w\frac{\partial Q_{i+1}(t)}{\partial DF(m_k)}\right]$$

Each quadratic is represented using the Lagrange basis

$$Q(t) = \sum_r \log DF(m_r)L_r(t)$$

Hence,

$$\frac{\partial Q(t)}{\partial DF(m_k)} = \frac{L_k(t)}{DF(m_k)}$$

provided that m_k is one of the three nodes used in the quadratic interpolation. Otherwise the derivative is zero.

3.6 Swap Curve Risk

For swap-curve nodes, discount factors are obtained through bootstrapping and are not directly implied by market quotes.

For calibration swap i,

$$F_i(DF_1, \ldots, DF_i, s_i) = 0$$

where s_i is the market par swap quote corresponding to maturity m_i.

Differentiating with respect to quote s_k,

$$\frac{\partial F_i}{\partial DF_i}\frac{\partial DF_i}{\partial s_k} + \sum_{j<i}\frac{\partial F_i}{\partial DF_j}\frac{\partial DF_j}{\partial s_k} + \frac{\partial F_i}{\partial s_k} = 0$$

Rearranging,

$$\frac{\partial DF_i}{\partial s_k} = -\frac{\sum_{j<i}\frac{\partial F_i}{\partial DF_j}\frac{\partial DF_j}{\partial s_k} + \frac{\partial F_i}{\partial s_k}}{\frac{\partial F_i}{\partial DF_i}}$$

Finally, the swap risk vector is obtained through

$$\frac{\partial PV}{\partial s_k} = \sum_i \frac{\partial PV}{\partial DF_i}\frac{\partial DF_i}{\partial s_k}$$

Although discount factors are obtained numerically through a bisection procedure, the sensitivities are not obtained by differentiating the root-finding iterations. Instead, the calibration equations themselves are differentiated analytically to obtain the required risk vectors.

4 System Architecture

4.1 Date Utilities

The `DateUtil` class converts maturity strings such as `1D`, `2W`, `3M`, and `5Y` into day counts. The implementation follows the problem convention that one week has 7 days, one month has 30 days, and one year has 360 days.

4.2 Interpolation Framework

The system uses an abstract interpolation interface:

`Interpolator`

Concrete implementations include:

- `LinearInterpolator`
- `AveragedQuadraticInterpolator`

This follows the Strategy Pattern. New interpolation methods, such as cubic spline interpolation, can be added by implementing the same interface without modifying the pricing or curve construction code.

4.3 Curve Construction

The `DiscountCurve` class stores calibrated discount factor nodes and provides access to interpolated discount factors.

Two curve builders are implemented:

- `CashCurveBuilder`
- `SwapCurveBuilder`

The cash curve builder directly converts market cash rates into discount factors.

The swap curve builder bootstraps discount factors sequentially from market par swap rates. For each swap maturity, the unknown discount factor is determined by solving the calibration equation

$$PV_{\text{fixed}} - PV_{\text{floating}} = 0$$

using a bisection root-finding algorithm. The calibrated discount factor is then added to the curve and used in the construction of subsequent maturities.

Since each swap calibration depends only on previously calibrated discount factors, the bootstrap procedure produces a lower triangular dependency structure which is later used in the analytical risk calculations.

4.4 Pricing Engine

The `SwapPricer` class calculates the present value and par swap rate of a new swap. The pricer depends only on the discount curve and interpolation method, and is independent of whether the curve was built from cash instruments or swap instruments.

4.5 Risk Engine

The `RiskCalculator` class computes analytical sensitivities. Separate functions are implemented for

- Cash curve with linear interpolation,
- Cash curve with averaged quadratic interpolation,
- Swap curve with linear interpolation,
- Swap curve with averaged quadratic interpolation.

The risk engine is separated from the pricing engine so that future risk measures can be added without modifying the pricing logic.

5 Extensibility

The system was designed to support future extensions with minimal changes.

- A new interpolation method can be added by deriving a new class from `Interpolator`.
- A new instrument type can be added by defining its cashflows and calibration equation.
- A new pricing model can reuse the existing `DiscountCurve` abstraction.
- Additional risk measures can be implemented inside the risk engine without changing curve construction or pricing logic.

This modular structure reduces coupling between components and improves maintainability.

6 Assumptions

The implementation uses the following assumptions:

- One week is treated as 7 days.
- One month is treated as 30 days.
- One year is treated as 360 days.
- Swaps used for curve calibration have semi-annual fixed and floating payments.
- Swaps with maturity below 6 months are treated as having a single exchange at maturity.
- The floating leg value is computed as $N(1 - DF(T))$.
- For the boundary case in averaged quadratic interpolation where two quadratics are not available, the available interpolation formula is used.

7 Validation

The implementation was validated using the following checks:

- Discount factors were checked for numerical reasonableness.
- Bootstrapped swap discount factors were verified by checking that the calibration swaps had approximately zero present value.
- Risk vectors were checked for consistency with the underlying interpolation and curve construction methodology.
- Linear and averaged quadratic interpolation outputs were compared for consistency.
- The output CSV format was matched to the required submission format.

8 Conclusion

The implemented system constructs discount curves from both cash and swap market instruments, supports multiple interpolation methods, prices a new swap, calculates par swap rates, and computes analytical risk vectors. The code is organized into modular components for date handling, interpolation, curve construction, pricing, and risk calculation. This structure makes the system extensible and suitable for adding future instruments or interpolation methods.